

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the
BACHELOR DEGREE IN COMPUTER SCIENCE

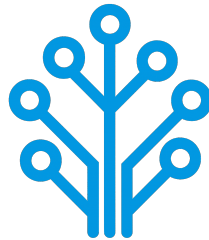
Field of Study : Software Engineering and Information Systems

Design and Implementation of a Network Intrusion Detection System

By

AMIRA BOUAFIF & HAJER JEMAA

Conducted within ITXTREE



Publicly defended on June 6, 2026 in front of the jury members:

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Academic Year: 2025-2026

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Authorization of graduation project report submission:

Professional Supervisor:

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Signature:

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Dedications

I dedicate this End-of-Study project

To my dear parents

Who have supported and encouraged me every step of the way. This is for you, in honor of all the sacrifices you have made for me. I hope to have made you both proud and happy.

To my dear sisters and brothers

Who have been a great support and a source of joy in my life. I am grateful for your love and encouragement throughout this journey.

To all my family and friends

Thank you all for your support and encouragement throughout all this time. I am truly blessed to have you in my life.

Amira BOUAFIF

Dedications

I dedicate this End-of-Study project

To my dear parents

To my dear sisters

To all my family and friends

Hajer JEMAA

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General Introduction

Chapter 1

Project Context

1.1 Introduction

This chapter serves as an introduction to the project. We begin with a presentation of the company hosting the internship, a study of existing network intrusion detection systems. Then, we present the problem and the proposed solution. Finally, we end the chapter with a conclusion.

1.2 Presentation of the company

ITXTREE is an early-stage technology startup based in Sidi Rzig, Tunisia, focused on building a solid foundation for future digital solutions it is built on the core values of integrity, quality, curiosity and adaptability.



1.3 Study of the existing

Cybersecurity is a highly sensitive domain where the ability to react quickly can determine the impact of an attack. The earlier a threat is identified, the more effectively its consequences can be limited.

For this reason, Network Intrusion Detection Systems (NIDS) play a central role in monitoring and analyzing network traffic. The most widely used approach relies on:

- **Signature-based:** These systems analyze network traffic, often through deep packet inspection and compares it against a signature database. When a match is found

an alert is triggered. Well-known tools such as Snort [1] and Suricata [2] are representative of this approach.

- **Anomaly-based:** Anomaly detection is learning normal behavior of the network through statistical analysis of traffic parameters. If there is a sudden increase of traffic on a port outside of normal thresholds, it might be an indication of a brand-new threat [1].

1.4 Criticism of the existing

Despite their widespread use, signature-based NIDS present several limitations:

- **High Maintenance and Reactivity:** Their effectiveness is essentially tied to the completeness and accuracy of their signature rule sets. As attack techniques continuously evolve, these systems require frequent updates to remain effective. This maintenance process is both time-consuming and reactive since new signatures can only be created after an attack has already been identified and analyzed.
- **Performance:** As the number of rules increases, the system must process a larger set of comparisons for each packet, which can lead to slower detection. In a context where reaction time is critical, this degradation directly affects the system's ability to respond effectively.
- **Rigidity and Evasion Vulnerability:** Attackers can exploit the rigidity of signature-based detection systems by introducing slight variations in attack patterns, allowing them to evade existing signatures while preserving the same malicious intent. Since many signature rules are publicly available or widely shared with the security community, attackers may analyze them to identify gaps and design variants that bypass detection mechanisms.

Overall, these systems rely on exact matching and predefined logics which limits their adaptability in the face of evolving and increasingly sophisticated threats.

1.5 Proposed solution

With the growing integration of Artificial Intelligence (AI) in security systems, new approaches aim to overcome the limits of traditional methods. To address these challenges, we propose an AI-driven NIDS capable of learning patterns from network traffic and classifying different types of network activity instead of relying solely on predefined rules.

1.6 Project Pipeline

A project pipeline is a structured sequence of stages that outlines the workflow and processes involved in the development and deployment of a project. It serves as a roadmap, guiding the team through various phases, from initial planning to final delivery.

Figure 1.1 illustrates the project pipeline for our network intrusion detection system.

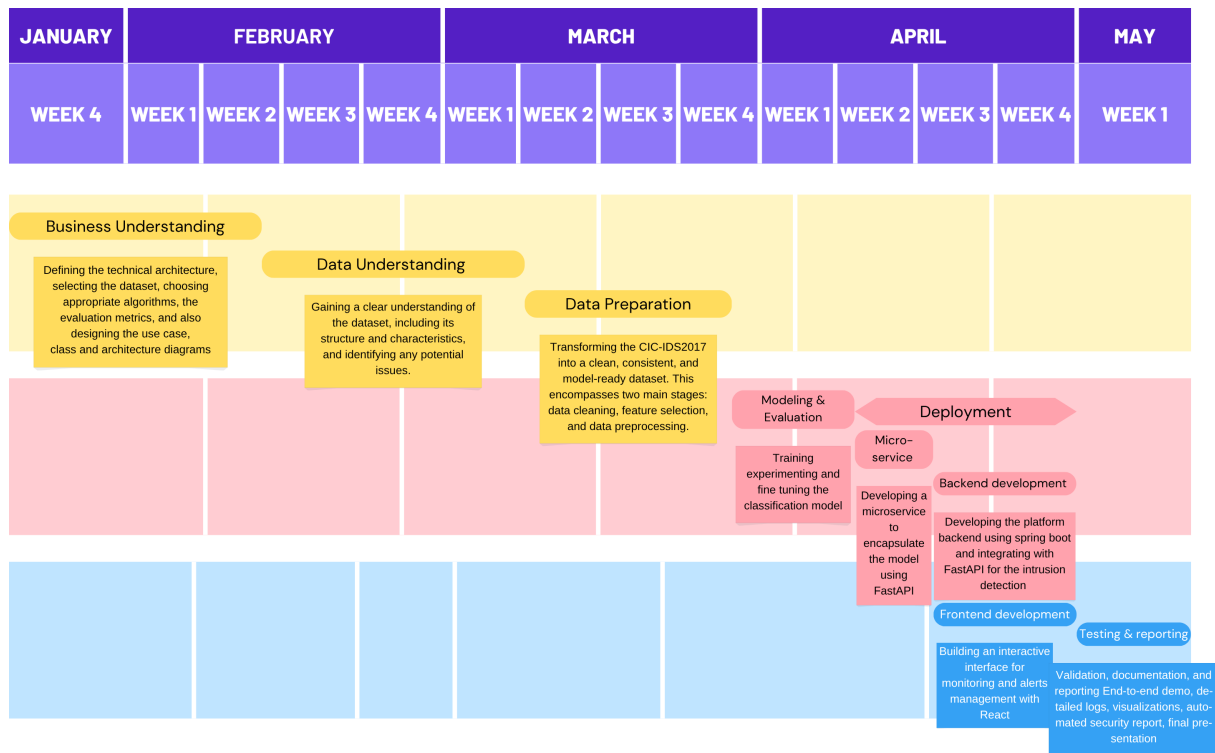


FIGURE 1.1 – Project Pipeline

1.7 Methodologies

In this section, we present the methodologies that we use in this project, including the machine learning methodology CRISP-DM and the software development methodology UML.

1.7.1 CRISP-DM

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology is a widely used framework for structuring machine learning projects. It consists of six phases:

- **Business Understanding:** Defining project objectives and requirements and gaining a clear understanding of the business problem.
- **Data Understanding:** Collecting and exploring the data to gain insights and assess its quality by identifying data quality issues.
- **Data Preparation:** Cleaning and preparing the data for modeling by selecting relevant features, handling missing values and splitting the dataset.
- **Modeling:** Selecting and applying appropriate modeling techniques, building, testing and tuning models to optimize their performance.
- **Evaluation:** Evaluating the model's performance and determining if it meets the business objectives.
- **Deployment:** Deploying the model into production, monitoring and maintaining the model's performance.

Figure 1.2 illustrates these phases and their relationships.

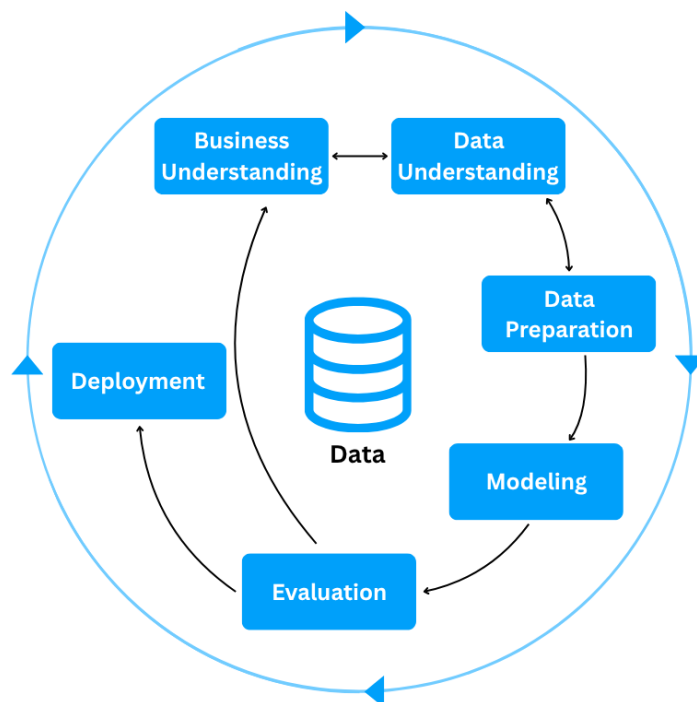


FIGURE 1.2 – CRISP-DM Methodology

1.7.2 UML

For the modeling of the software system, we use UML (Unified Modeling Language), a standardized modeling language used to specify, visualize, construct, and document the artifacts of software systems through diagrams such as use case, class, and sequence diagrams. There are two types of UML diagrams:

- **Behavioral Diagrams:** State machine, activity, use case, sequence, communication and interaction overview diagrams.
- **Structural Diagrams:** Class, object, component, deployment and package diagrams.

UML makes complex systems easier to design and communicate, it provides a common language for developers, architects, and stakeholders [3].

1.8 Conclusion

In this chapter, we introduce the host company ITXTREE, study existing network intrusion detection systems, discuss their shortcomings and finally present a solution. In the next chapter we specify the system requirements.

Chapter 2

Requirements Specification

2.1 Introduction

Throughout this chapter, we specify both the functional and non-functional requirements, identify the system actors, and present the use case diagram. By defining these elements, we aim to establish a clear and shared understanding of the system's expected behavior and constraints, providing a solid foundation for the subsequent design and implementation phases.

2.2 Functional Requirements

Functional requirements define *what* the system should perform. The main functionalities of our system are as follows:

- Authenticate;
- View Dashboard;
- View Statistics summary;
- Trigger Network Analysis;
- View Alerts List;
- View Reports;

2.3 Non-Functional Requirements

Non-functional requirements define *how* the system should perform its functions. They describe the quality attributes and constraints of the system:

- **Performance:** The system must process network traffic and generate analysis results with minimal latency, ensuring fast detection of potential intrusions.
- **Security:** The system must ensure secure authentication and enforce role-based access control, preventing unauthorized actions and protecting system integrity, while supporting non-repudiation to guarantee accountability and traceability of all operations.

- **Maintainability:** The system should be easy to maintain, allowing for efficient debugging, updates, and modifications to existing components.
- **Usability:** The system should provide an intuitive interface, enabling users to efficiently navigate and interact with its functionalities.

2.4 Actors Identification

Actors are external entities, users, organizations, or external systems that interact with the system to achieve one or more functional requirements, with each actor representing a specific role.

In the context of our project, we identify two primary actors, as illustrated in Figure 2.1

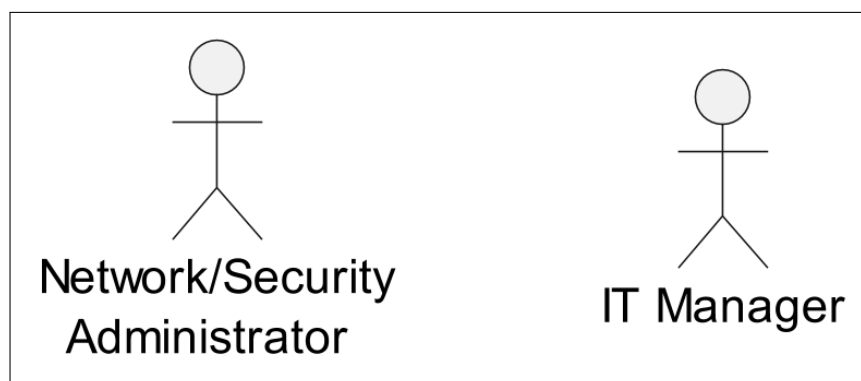


FIGURE 2.1 – Actors

Table 2.1 presents the roles of each actor in our system.

TABLE 2.1 – Roles of Actors

Actor	Role
Network/Security Administrator	- Authenticate - View Dashboard - View Statistics Summary - Trigger Network Analysis - View Alerts List
It Manager	- Authenticate - View Dashboard - View Statistics Summary - View Reports

2.5 Use Case Diagram

A use case diagram is a visual representation of *how* users interact with the system and *what* a system does through different use cases. Figure 2.2 illustrates that diagram.

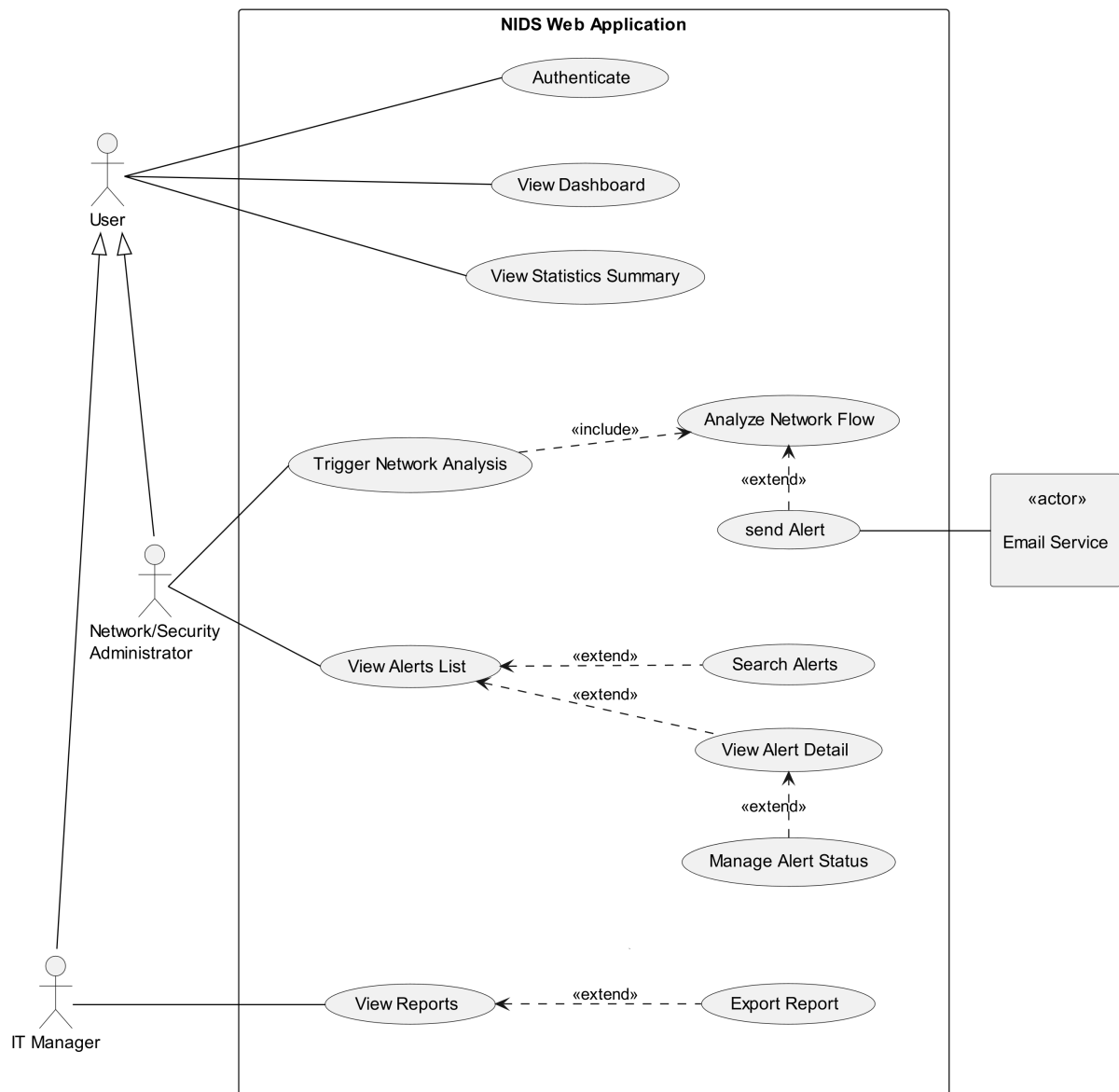


FIGURE 2.2 – Use Case Diagram

2.6 Conclusion

In this chapter, we outline the functional and non-functional requirements of our system, identify the primary actors, and represent the way they interact with the system and the system's functionality via a use case diagram. In the next chapter, we present the conceptual design of the platform.

Chapter 3

Conceptual Design

3.1 Introduction

3.2 Sequence Diagrams

3.3 Class Diagram

3.4 Deployment Diagram

3.5 Conclusion

Chapter 4

Data Collection and Preparation

4.1 Introduction

4.2 Data Collection

4.3 Data Understanding

The Data Understanding phase of the CRISP-DM methodology focuses on loading, describing, and analyzing the dataset to build a solid foundation for the rest of the project. This section dives into the four tasks of this phase: **Initial Exploration**, **Data Description**, **Data Exploration**, and **Data Quality**, with each task targeting an aspect of the data, from its structure and content to its quality.

4.3.1 Initial Exploration

The CIC-IDS2017 dataset is captured over five days, it is distributed across 8 files split by day and session: **Monday**, **Tuesday**, and **Wednesday** each corresponding to a single file, while **Thursday** and **Friday** are split into AM and PM sessions. The **Friday PM** session is further divided into two files to isolate the two attacks, **DDoS** and **PortScan**.

4.3.2 Data Description

As mentioned in the previous subsection, the dataset is split into 8 files as shown in Table 4.1.

TABLE 4.1 – Dataset Files

File	Number of Rows	Number of Columns
Monday	529,918	79
Tuesday	445,909	79
Wednesday	692,703	79
Thursday AM	170,366	79
Thursday PM	288,602	79
Friday AM	191,033	79
Friday PM (DDoS)	225,745	79
Friday PM (PortScan)	286,467	79
Total	2,830,743	—

The dataset contains a total of 2,830,743 rows across the 8 files. All files have consistent columns, which is expected as they are generated by the same CICFlowMeter pipeline. This consistency allows for seamless concatenation of the files into a single DataFrame for the subsequent analysis.

- **Whitespace Issue:** Leading white spaces is found in columns names and are stripped across all files as they can raise `KeyError` exceptions when accessing columns by name.
- **Corrupted Labels:** Corrupted labels are discovered in the Thursday AM file, where three labels (`Web Attack ? Brute Force`, `Web Attack ? XSS`, `Web Attack ? Sql Injection`) contain mojibake character instead of en dash (-), and are therefore corrected.

These fixes are purely cosmetic and do not alter the underlying data values.

- **Feature Description:** The dataset contains 79 columns, of them 78 are numeric level-flow features and the last one is the target Label. These features can be grouped into 12 categories as shown in Table 4.2.

TABLE 4.2 – Feature Groups

Group	Features	Description
Flow Metadata	Destination Port, Flow Duration	Total duration of the flow and the destination port
Packet/Byte Count	Total Fwd Packets, Total Backward Packets, Total Length of Fwd Packets, Total Length of Bwd Packets, act_data_pkt_fwd	Number of packets and packet bytes in both forward (from source to destination) and backward directions (from destination to source), and packets having actual data in the forward direction.
Packet/Segment Length Statistics	Fwd Packet Length Max/Min/Mean/Std, Bwd Packet Length Max/Min/Mean/Std, Max/Min Packet Length, Packet Length Mean/Std/Variance, Average Packet Size, Avg Fwd Segment Size, Avg Bwd Segment Size	Packets and segments length statistics (e.g. max, min, mean, std etc...).
Flow Rate	Flow Bytes/s, Flow Packets/s, Fwd Packets/s, Bwd Packets/s, Down/Up Ratio	Flow throughput, forward and backward packet rates, flow asymmetry ratio (backward packets (download) to forward packets (upload)).
Inter-Arrival Time	Flow IAT Max/Min/Mean/Std, Fwd IAT Max/Min/Mean/Std/Total, Bwd IAT Max/Min/Mean/Std/Total	Time between consecutive packets in the flow, forward, and backward directions.
TCP Flag Count	FIN Flag Count, SYN Flag Count, RST Flag Count, PSH Flag Count, ACK Flag Count, URG Flag Count, CWE Flag Count, ECE Flag Count, Fwd PSH Flags, Bwd PSH Flags, Fwd URG Flags, Bwd URG Flags	TCP flags counts for the entire flow, forward and backward directions.
Header Length	Fwd Header Length, Bwd Header Length, Fwd Header Length.1	Backward and forward header lengths.
Bulk Transfer Statistics	Fwd Avg Bytes/Bulk, Fwd Avg Packets/Bulk, Fwd Avg Bulk Rate, Bwd Avg Bytes/Bulk, Bwd Avg Packets/Bulk, Bwd Avg Bulk Rate	Average forward and backward bulk (a sequence of consecutive packets in the same direction) transfer statistics.

Subflow Packet/Byte Count	Subflow Fwd Packets, Subflow Fwd Bytes, Subflow Bwd Packets, Subflow Bwd Bytes	Number of packets and packet bytes in both directions in a sub-flow.
Window/Segment Size	Init_Win_bytes_forward, Init_Win_bytes_backward, min_seg_size_forward	Window sizes in both the forward and backward directions and the minimal segment size in the forward direction.
Active/Idle Time	Active Max/Min/Mean/Std, Idle Max/Min/Mean/Std	Duration of active and idle periods within a flow.
Target	Label	The target label indicating the class the flow belongs to.

- **Duplicate Column:** The column `Fwd Header Length.1` is identified as a duplicate of `Fwd Header Length`, confirmed by the fact that both contain the same values.
- **Packet Length Mean and Average Packet Size** both measure the average packet size, but the two do not hold the same values as they are calculated differently: the former uses the statistical mean, while the latter divides the total length of packets by the total number of packets[4].

All 54 features are of type `int64`, 24 features are of type `float64`, with the last feature, the target *Label*, being of type `object`.

Table 4.3 presents the statistics of some of the features.

TABLE 4.3 – Feature Statistics

Feature	Min	Max	Mean	Std
Flow Duration	-13	119,999,998	14,785,663.92	33,653,744.09
Total Backward Packets	0	121,922	10.39	997.39
act_data_pkt_fwd	0	213,557	5.42	636.43
Min Packet Length	0	1,448	16.43	25.24
Flow Bytes/s	-241,000,000	inf	inf	inf
Flow IAT Max	-13	120,000,000	9,182,475.32	24,459,539.25
Bwd PSH Flags	0	0	0	0
Fwd Header Length	-322,212,234,632	4,644,908	-25,997.39	21,052,857.87
Bwd Avg Bulk Rate	0	0	0	0

Some of the feature has negative, inf and constant statistic of zero. Those features are further investigated in the data quality section.

- **Class Distribution:** The dataset contains 15 classes, BENIGN and 14 attacks, Table 4.4 shows the distribution of these classes across the dataset files.

TABLE 4.4 – Class Distribution

File	Class	Class Count
Monday	BENIGN	529,918
Tuesday	BENIGN	432,074
	FTP-Patator	7,938
	SSH-Patator	5,897
Wednesday	BENIGN	440,031
	DoS Hulk	232,443
	DoS GoldenEye	10,293
	DoS Slowloris	5,796
	DoS Slowhttptest	5,499
	Heartbleed	11
Thursday AM	BENIGN	168,186
	Web Attack - Brute Force	1,507
	Web Attack - XSS	652
	Web Attack - Sql Injection	21
Thursday PM	BENIGN	288,566
	Infiltration	36
Friday AM	BENIGN	189,067
	Bot	1,966
Friday PM (DDoS)	DDoS	128,027
	BENIGN	97,718
Friday PM (PortScan)	PortScan	158,930
	BENIGN	127,537

With the exception of Monday, which contains only BENIGN traffic, all files contain at least one attack class alongside BENIGN samples. Each attack class is present in only one file.

Table 4.5 provides a description of the 14 attack classes of the dataset.

TABLE 4.5 – Attack Classes

Class	Description
FTP-Patator (File Transfer Protocol Patator)	A brute force attack using the Patator tool targeting FTP servers, an old unencrypted protocol that transmits data and credentials in plain text, using trial and error to guess credentials in order to gain unauthorized access.
SSH-Patator (Secure Shell Patator)	A brute force attack using the Patator tool targeting SSH servers, a secure encrypted protocol that encrypts the entire session including the login, using trial and error to gain unauthorized access to individual accounts.

DoS Hulk	A Denial-of-Service attack using the HULK tool, which is used to overwhelm web servers by generating a large number of unique randomized requests at irregular intervals.
DoS GoldenEye	A Denial-of-Service attack using the GoldenEye tool, which dynamically generates multiple parallel connections (multi-threaded) to a target URL while keeping them alive to exhaust the available connection slots, simultaneously forcing the server to bypass caching mechanisms and fully process every request.
DoS slowloris	A Denial-of-Service attack using the Slowloris tool, which enables a single machine to overwhelm a web server by opening multiple connections and keeping them open by sending partial requests (only headers) periodically, while requiring very low bandwidth, hence its name, as it operates in a low and slow manner.
DoS Slowhttpstest	A Denial-of-Service attack using the SlowHTTPTest tool, which creates very slow connections by sending seemingly legitimate requests using low bandwidth and intentionally delaying their completion, thereby consuming the limited number of concurrent connections the server can handle.
Heartbleed	An attack in which the attacker exploits the Heartbleed bug in the OpenSSL cryptographic library to read sensitive memory contents by sending malformed heartbeat requests demanding more memory than the actual payload from the server running the vulnerable version of OpenSSL, after which the server allocates the requested memory without any verification and returns it, exposing whatever sensitive data happened to be stored in that memory chunk at that moment. It is used to steal private encryption keys, usernames and passwords, and to intercept session tokens, permitting session hijacking by providing the stolen token to the server so that the attacker is treated as a legitimate logged-in user, as well as to decrypt previously captured encrypted traffic that the attacker had intercepted beforehand.
Web Attack - Brute Force	An attack that uses a forceful approach to steal credentials to gain unauthorized access to systems. It can be carried out in many ways, from trying common credential combinations manually, to using dictionary attacks, or a combination of both.

Web Attack - XSS	An attack in which malicious scripts are injected into trusted websites, by exploiting their input validation vulnerability, and executed by other end users' browsers, which have no way to distinguish it from legitimate content.
Web Attack - Sql Injection	An attack in which the attacker interferes with the queries that an application makes to its database enabling him to view data that he is not supposed to have access to, including data belonging to other users and any other data that the platform has access to, as well as modify or delete it. It is carried out by injecting malicious SQL queries into an input field, which the vulnerable application passes directly to the database as a query instead of treating it as plain data. The database then executes it, allowing the attacker to manipulate the database in unintended ways.
Infiltration	A multi-stage attack to gain unauthorized access to systems, in which the attacker first deploys a malicious file disguised as a legitimate one, such as a document or a downloadable link, containing a backdoor. The victim executes the file unknowingly, after which the backdoor establishes a reverse connection back to the attacker's machine, granting the attacker remote control over the compromised system. Once inside, the attacker scans the victim's internal network using tools (e.g. Nmap) to discover live hosts (any active and reachable device on the network), as well as open ports and running services. This allows the attacker to map the internal network and identify potential targets for further infiltration. The attacker then moves laterally by infiltrating another machine, spreading the attack further across the network. Finally, an attempt is made to exfiltrate sensitive data (transferring confidential information out of the network), and to escalate privileges on the compromised systems in order to gain access to even more restricted and sensitive resources.
Bot	An attack carried out using the ARES botnet, a malware-based network of compromised computers built on a RAT (Remote Access Trojan) and botnet framework, which allows the attacker to take control of infected machines and coordinate them for malicious purposes. It first infects victim machines using phishing emails or malicious downloads, after which the infected machines connect back to the attacker's Command and Control (C&C) server. Finally, the attacker issues commands to all bots, that is, all infected machines, simultaneously.

DDoS	An attack used to overwhelm the target system with a flood of traffic from multiple sources, making it unable to respond to legitimate requests, potentially leading to a complete shutdown of the system.
PortScan	An attack used to discover open ports and whether they are receiving or sending data, in addition to potential vulnerabilities that could be exploited.

4.3.3 Data Exploration

4.4 Data Preparation

4.5 Data Augmentation

4.6 Conclusion

Chapter 5

Modeling

5.1 Introduction

5.2 State of the Art

5.3 Adopted Approach

5.4 Training and Hyperparameter Optimization

5.5 Conclusion

Chapter 6

Realization

6.1 Introduction

6.2 Development Tools and Technologies

6.3 Performance Analysis and Evaluation Metrics

6.4 User Interfaces

6.5 Integration

6.6 Conclusion

General Conclusion

Webography

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